

Qatar-1b

R-2772

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160805 · created 2026-09-03T23:15:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457605.84662 +/- 0.00086 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.005 +/- 0.055
Transit depth (Rp/Rs)^2	0.0021 +/- 0.05 [%]
Orbital Inclination (inc)	82.07 +/- 2.86
Airmass coefficient 1 (a1)	1.01 +/- 0.6
Airmass coefficient 2 (a2)	0.08 +/- 0.85
Scatter in the residuals of the lightcurve fit is	66.65 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	5.5
Transit Duration (day)	0.037 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.005 ± 0.055 vs 0.1463 (deviation 2.57σ)
Duration fit vs published	0.037 d vs 0.0692 d (deviation 1.19σ)
Mid-time O–C	-0.2 min
Depth SNR	0.1
Residual scatter	66.65 %

Light curve

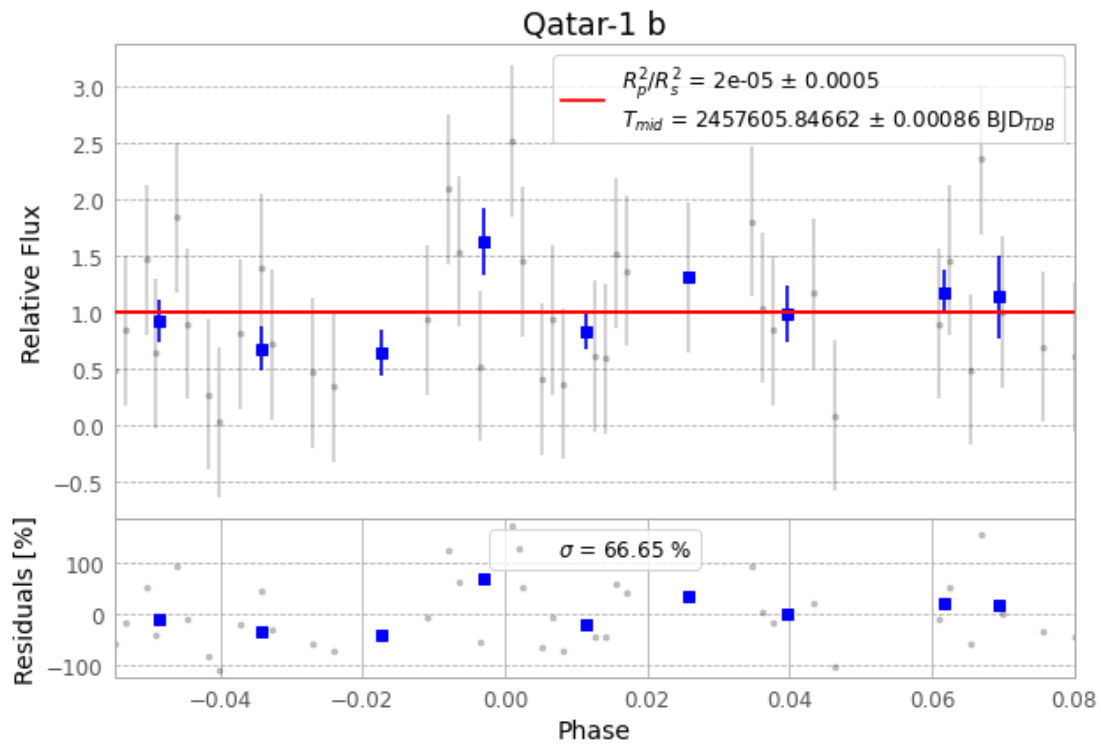


Figure 1 — FinalLightCurve_Qatar-1 b_2016-08-04.png (SHA-256 f91caeacd0c0e54f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [102, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 993ae046515b23e6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

96 FITS frames (63 MB), Qatar-1160805062413.FITS → Qatar-1160805110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160805.log (048ecf3dd83f...), FinalLightCurve_Qatar-1 b_2016-08-04.png (f91caeacd0c0...), FinalLightCurve_Qatar-1 b_2016-08-04.csv (bf14dee62f33...)

Compute resources

Wall time 161.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:15Z.